

# PULSE POS

## Detailed Version Report

Version 0.0.0 (Pre-release)

Report Date: January 7, 2026

Last Code Update: January 14, 2025

Point-of-Sale & Inventory Management System

Electron + React + RxDB + Supabase

# 1. Executive Summary

Pulse POS is a comprehensive Point-of-Sale and Inventory Management system designed for retail businesses. It is built as an Electron-based desktop application with offline-first capabilities and cloud sync support. The system targets Bulgarian retail markets (default currency: BGN) with multi-language support.

## Key Highlights

- 7 of 13 development phases completed (53.8%)
- 16 major feature modules implemented
- 40+ granular permissions for role-based access control
- 3 languages supported: English, Bulgarian, Turkish
- Offline-first architecture with RxDB + Supabase sync

# 2. Technology Stack

## Frontend (Desktop Client)

| Technology    | Version | Purpose              |
|---------------|---------|----------------------|
| React         | 18.2.0  | UI Framework         |
| TypeScript    | 5.2.2   | Type Safety          |
| Vite          | 5.1.6   | Build Tool           |
| Electron      | 30.0.1  | Desktop Runtime      |
| TailwindCSS   | 3.4.1   | Styling              |
| Zustand       | 4.5.0   | State Management     |
| React Query   | 5.0.0   | Server State         |
| i18next       | 25.6.3  | Internationalization |
| Framer Motion | 12.24.7 | Animations           |

## Data Layer

| Technology | Version    | Purpose                |
|------------|------------|------------------------|
| RxDB       | 16.21.1    | Offline-first Database |
| Dexie      | integrated | IndexedDB Wrapper      |
| Supabase   | 2.39.3     | Cloud Backend/Sync     |
| Zod        | 3.22.4     | Schema Validation      |

## Reporting & Documents

| Library        | Purpose            |
|----------------|--------------------|
| Recharts 3.4.1 | Data Visualization |
| jsPDF 3.0.4    | PDF Generation     |
| xlsx 0.18.5    | Excel Export       |

jsbarcode 3.12.1  
qrcode.react 4.2.0  
html2canvas 1.4.1

Barcode Generation  
QR Code Generation  
Chart Screenshots

### 3. Project Architecture

#### Monorepo Structure

- pulse-monorepo/
  - apps/
    - desktop-client/ - Electron + React Desktop App
    - src/features/ - Feature-based modules (16 modules)
    - src/layouts/ - Sidebar, layouts
    - src/components/ - Shared components
    - src/hooks/ - Custom React hooks
    - src/utills/ - Utility functions
    - src/locales/ - i18n translations (en, bg, tr)
  - packages/
    - core-logic/ - Shared Business Logic
    - database/ - RxDB, Dexie, Supabase clients
    - services/ - Business services (12 services)
    - store/ - Zustand stores (auth, cart)
    - types/ - Zod schemas & TypeScript types

#### Feature Modules (16 Total)

| Module           | Description                    |
|------------------|--------------------------------|
| analytics        | Charts, reports, business inte |
| auth             | Login, authentication          |
| customer-display | Customer-facing screen         |
| customers        | CRM, customer profiles         |
| dashboard        | Quick actions, overview        |
| employees        | User, role, time clock managem |
| inventory        | Stock management, adjustments  |
| layaway          | Layaway order system           |
| locations        | Multi-location, stock transfer |
| pos              | Point-of-sale terminal         |
| promotions       | Discount campaigns             |
| purchase-orders  | Supplier ordering              |
| receipts         | Receipt designer, Z-reports, I |
| settings         | App configuration              |
| suppliers        | Supplier management            |

## 4. Database Schema

The system uses 30+ database tables organized into logical groups:

### **Sales & Transactions**

- sales - Completed transactions
- sale\_items - Line items per sale
- shifts - Cash drawer shifts
- cash\_transactions - Pay in/out records
- parked\_sales - Suspended transactions

### **Products & Inventory**

- products - Product catalog
- product\_barcode - Multi-barcode support
- stock\_movements - Stock audit trail
- categories - Product categories

### **Customers & Loyalty**

- customers - Profiles with tiers (Bronze/Silver/Gold/Platinum)
- credit\_transactions - Store credit ledger

### **Suppliers & Purchasing**

- suppliers - Supplier directory
- product\_suppliers - Product-supplier links
- purchase\_orders - PO headers
- purchase\_order\_items - PO line items

### **Payments & Financial**

- gift\_cards - Gift card registry
- layaway\_orders - Layaway contracts
- layaway\_payments - Layaway payment schedule

### **Multi-Location**

- locations - Store/branch registry
- stock\_transfers - Inter-location transfers
- location\_pricing - Location-specific pricing

### **Employees & RBAC**

- users - Employee accounts
- roles - Permission roles
- permissions - System permissions
- time\_clock\_entries - Clock in/out records
- activity\_logs - Audit trail

## 5. Core Services

12 business logic services power the application:

| Service              | Purpose                        |
|----------------------|--------------------------------|
| AuthService          | Authentication, password hashi |
| ActivityLogService   | Audit trail logging            |
| GiftCardService      | Gift card operations           |
| LayawayService       | Layaway order management       |
| LocationService      | Multi-location CRUD            |
| LoyaltyService       | Points, tiers, rewards         |
| MarketService        | Market mode operations         |
| PurchaseOrderService | PO workflow                    |
| StockTransferService | Inter-location transfers       |
| SupplierService      | Supplier management            |
| syncService          | Cloud synchronization          |
| TimeClockService     | Employee time tracking         |

## 6. Feature Implementation Status

### Phase 1: Analytics & Business Intelligence [COMPLETE]

- RevenueTrendsChart - Line/bar charts with payment breakdown
- TopProductsChart - Top 10 by revenue/quantity
- ProfitMarginAnalysis - Margin health indicators
- SalesHeatmap - Hour x Day visualization
- CashierPerformance - Employee metrics
- LowStockAlerts - Real-time stock monitoring
- ReportExporter - PDF/Excel export

### Phase 2: Enhanced CRM & Loyalty [COMPLETE]

- Customer profiles with extended fields
- Tier System: Bronze -> Silver -> Gold -> Platinum
- Automatic point accumulation (1 point per 10 BGN)
- Visit tracking, tags, preferences, notes
- Referral code support

### **Phase 3: Advanced Inventory & Suppliers [COMPLETE]**

- Full supplier directory with contacts
- Product-supplier relationships
- Purchase Order System with workflow states
- Stock adjustment with reason codes
- Adjustment history (30 days)

### **Phase 4: Payment & Financial Features [COMPLETE]**

- Gift Cards: Sell, redeem, reload, track balance
- Store Credit: Issue, redeem, transaction history
- Layaway System: Orders, payments, restocking fees

### **Phase 5: Multi-Location Support [COMPLETE]**

- Location CRUD with manager assignment
- Stock Transfers: Request -> Approve -> Ship -> Receive
- Multi-location dashboard with comparisons
- P&L Report by location

### **Phase 6: Employee Management & RBAC [COMPLETE]**

- 4 Default Roles: Admin, Manager, Cashier, Stock Clerk
- 40+ Permissions across 8 categories
- PIN-based quick login
- Time clock with break tracking
- Activity log viewer with filters
- RequirePermission component for UI enforcement

### **Phase 7: Receipt & Document Enhancements [COMPLETE]**

- Receipt Designer: Drag-and-drop template builder
- Component types: Text, Logo, Fields, Items, Totals, Barcode, QR
- Z-Report: End-of-day fiscal summary
- Label Printing: Product labels with barcodes
- Multiple size presets (38x25mm to 80x50mm)

### **Phases 8-13: NOT STARTED**

- Phase 8: Returns & Exchanges
- Phase 9: Distributor/Wholesale Features
- Phase 10: Tax & Compliance
- Phase 11: Mobile App Development
- Phase 12: Integrations & API
- Phase 13: Security & Audit

## 7. UI/UX Features

### Navigation & Layout

- Collapsible sidebar with icon + label
- 20+ navigation items organized by category
- Permission-controlled menu visibility
- Dark/Light theme toggle
- Responsive design

### POS Terminal

- Product grid with favorites
- Barcode scanner support (useBarcodeScanner hook)
- Cart with quantity controls
- Price override (with permission)
- Discount application
- Customer attachment
- Parked sales & Pay-it-forward feature
- Split payment support
- Keyboard shortcuts (F1-F5)

### Customer Display

- Secondary window for customer-facing screen
- Real-time cart display
- Thank you / Welcome messages

## 8. Internationalization

### Supported Languages

| Code | Language  | Status               |
|------|-----------|----------------------|
| en   | English   | Complete (~550 keys) |
| bg   | Bulgarian | Complete             |
| tr   | Turkish   | Complete             |

### Translation Categories

Common, Shift, Customer Display, Customer, Sidebar, Analytics, POS, Cart, Payment, Receipt, Inventory, Dashboard, Settings, Promotions, Auth, Cashiers, Locations

## 9. Notification System

### Desktop Notifications

- Low stock alerts (configurable threshold)
- High-value sale alerts (configurable amount)



- Failed sync notifications

## **In-App Notifications**

- Sonner toast library
- Success, Warning, Error, Info types
- Auto-dismiss with custom duration
- Settings UI for enable/disable per category

## 10. Known Issues & Technical Debt

- PromotionsScreen.tsx - Type mismatch with bundle/tiered promotion types
- MarketService.ts - Boolean comparison warnings (is\_active === 1)
- Missing translation keys for some new features
- Email/SMS receipts - Pending service integration
- Sound effects - Audio files not yet added

## 11. Recommended Next Steps

### High Priority

- Phase 8: Returns & Exchanges - Return workflow, exchange processing, refund methods
- Phase 10: Tax & Compliance - Multi-tax rate support, Bulgarian NRA fiscal receipts

### Medium Priority

- Phase 9: Distributor/Wholesale - B2B customer types, volume pricing, credit terms

### Lower Priority

- Phase 11-13: Mobile app, API integrations, and security enhancements

## 12. Git Status Summary

Branch: main

### Modified Files (41 files)

- Core app files (App.tsx, Sidebar, etc.)
- All major feature screens updated
- Localization files expanded
- Core-logic services enhanced

### New Files (17 untracked)

- QuickActionsDashboard.tsx
- StockAdjustmentModal.tsx
- ShortcutsHelpModal.tsx
- useBarcodeScanner.ts hook
- Receipt feature module
- Settings store
- Test infrastructure

## 13. Performance & Security

## **Performance Considerations**

- Offline-first: RxDB with IndexedDB for instant local operations
- Lazy loading: Feature modules loaded on demand
- Optimistic updates: UI updates before server confirmation
- Efficient queries: RxDB reactive queries with subscription cleanup

## **Security Features**

- PIN-based authentication
- Role-based access control (RBAC)
- Activity logging for audit trails
- Session timeout detection
- Permission-gated UI components

# Summary

This report provides a comprehensive overview of the Pulse POS project at its current state. The system has completed 7 major phases with robust POS, inventory, CRM, and employee management capabilities, while 6 phases remain for future development.

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**Progress: 7/13 Phases Complete (53.8%)**

16 Feature Modules | 12 Services | 30+ Database Tables | 40+ Permissions

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Built with Electron + React + TypeScript + RxDB + Supabase  
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